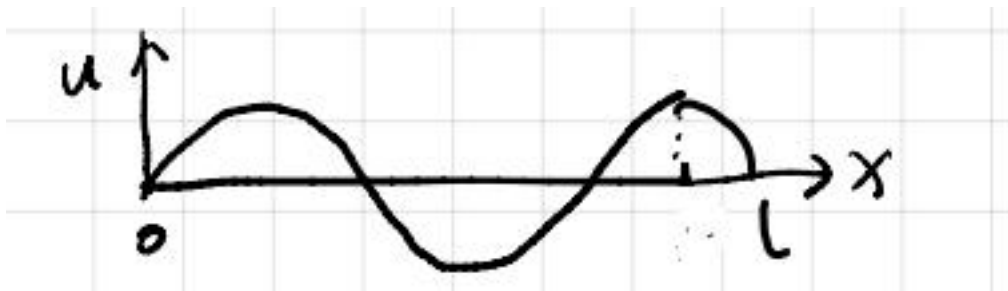


Chapter 9

Separation of variables

S13.1

Given a string, both ends fixed. $u(x, t)$ represents the position of the string,



We know that wave equation is

$$u_{tt}^2 - a^2 u_{xx} = 0$$

The edge condition is

$$u|_{x=0} = 0 \quad u|_{x=l} = 0$$

The initial condition is

$$u|_{t=0} = \phi(x)$$

The initial velocity is

$$u_t|_{t=0} = \psi(x)$$

"两端固定的自由弦振动"

Step 1

$$\begin{aligned}u(X, t) &= X(x)T(t) \\ T''(t)X(x) - a^2X''(x)T(t) &= 0 \\ \frac{1}{a^2} \cdot \frac{T''(t)}{T(t)} &= \frac{X''(x)}{X(x)}\end{aligned}$$

The only way to let them equal is

$$\begin{aligned}\frac{1}{a^2} \cdot \frac{T''(t)}{T(t)} &= \frac{X''(x)}{X(x)} = -\lambda \\ X''(x) + \lambda X(x) &= 0 \\ T''(t) + \lambda a^2 T(t) &= 0 \\ X(x)T(t)|_{x=0} &= X(0)T(t) = 0 \Rightarrow X(0) = 0 \\ X(x)T(t)|_{x=1} &= X(l)T(t) = 0 \Rightarrow x(l) = 0\end{aligned}$$

Then we can get a linear second order differential condition.

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$$

When $\lambda = 0$ $x''(x) = 0$ $x(t) = Ax + B$

We know $A = B = 0$

$$\Rightarrow X(x) \equiv 0$$

$$u(x, t) = X(x)T(t) = 0$$

which is not true.

When $\lambda < 0$.

$$\begin{aligned}X(x) &= Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{\lambda}x} \\ X(0) &= A + B = 0 \\ X(l) &= Ae^{2l} + Be^{-2l} = 0\end{aligned}$$

According to

$$\begin{vmatrix} 1 & 1 \\ e^{2l} & e^{-2l} \end{vmatrix} \neq 0$$

Then A and B must be zero.

So $X(x) \equiv 0$, which is not true.

Then λ has to be positive.

Step 2

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$$

This is an eigenvalue problem.

$$\begin{aligned} X(x) &= A \cos \sqrt{\lambda} x + B \sin \sqrt{\lambda} x \\ X(0) &= A \cdot 1 + B \cdot 0 = 0 \Rightarrow A = 0 \\ X(l) &= B \sin \sqrt{\lambda} \cdot l = 0 \quad B \neq 0 \end{aligned}$$

So $\sin \sqrt{\lambda} \cdot l = 0$ must be zero.

$$\sqrt{\lambda} \cdot l = n\pi \quad n = 1, 2, 3, \dots$$

$X_n(x) = \sin \frac{n\pi}{l} x$ is eigen function.

λ is eigenvalue.

$$\sqrt{\lambda} = \frac{n\pi}{l} \quad n = 1, 2, 3, \dots$$

Step 3

$$\begin{aligned} T''(t) + \lambda a^2 T(t) &= 0 \\ T_n &= C_n \cos \sqrt{\lambda} at + D_n \sin \sqrt{\lambda} at \\ &= C_n \cos \frac{n\pi}{2} at + D_n \sin \frac{n\pi}{2} at \quad n = 1, 2, 3, \dots \\ u_n(x, t) &= \sin \frac{n\pi}{l} x \times \left(C_n \cos \frac{n\pi}{2} at + D_n \sin \frac{n\pi}{2} at \right) \end{aligned}$$

After that, we can get

$$\begin{aligned} (u_n)_{tt} - a^2 (u_n)_{xx} &= T_n''(t) X_n(x) - a^2 T_n(t) X_n''(x) \\ &= -\lambda a^2 T_n(t) X_n(x) - a^2 T_n(t) [-\lambda X_n(x)] \\ &= 0 \end{aligned}$$

Step 4

For $u_n(x, t)$,

$$(u_n)_{tt} - a^2(u_n)_{xx} = 0$$
$$u|_{x=0} = u|_{x=l} = 0$$

Then,

$$u_n(x, t) = \sin \frac{n\pi}{t} \times \left(C_n \cos \frac{n\pi}{t} at + D_n \sin \frac{n\pi}{t} at \right)$$
$$u_n|_{t=0} = \phi_n(x) \Rightarrow C_n \sin \left(\frac{n\pi}{t} x \right) = \phi_n(x)$$

According to Fourier Transform, we can get

$$\sum_{n=1}^{\infty} u_n(x, t) = u(x, t)$$
$$u(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi}{t} \times \left(C_n \cos \frac{n\pi}{t} at + D_n \sin \frac{n\pi}{t} at \right)$$
$$\phi(x) = \sum_{k=1}^{\infty} D_n \sin \frac{n\pi}{t} x$$
$$D_n = \frac{2}{10} \int_0^{T_0} \phi(x) \sin \frac{n\pi}{t} \times dx = \frac{1}{t} \int_0^{2l} \phi(x) \sin \frac{n\pi}{t} \times dx$$
$$u_t|_{t=0} = \psi(x)$$
$$u_t(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi}{l} x \left(C_n \frac{n\pi a}{l} \cos \frac{n\pi}{l} at + D_n \left(-\frac{n\pi a}{l} \right) \sin \frac{n\pi}{l} at \right)$$
$$\psi(x) = u_t|_{t=0} = \sum_{n=1}^{\infty} \sin \left(\frac{n\pi}{l} x \right) \cdot C_n \frac{n\pi}{l}$$
$$C_n \frac{n\pi a}{l} = \frac{2}{T_0} \int_0^{T_0} \psi(x) \sin \frac{n\pi}{l} x dx$$
$$C_n = \frac{1}{n\pi a} \int_0^{2l} \psi(x) \sin \frac{n\pi}{l} x dx$$
$$u(x, t) = \sum_{n=1}^{\infty} \sin \left(\frac{n\pi}{l} x \right) \left[C_n \sin \frac{n\pi}{l} t + D_n \cos \frac{n\pi}{l} t \right]$$

S13.3

$$\begin{cases} u_{xx} + u_{yy} = 0 & 0 < x < a \quad 0 < y < b \\ u|_{x=0} = 0 & \frac{\partial u}{\partial x}\bigg|_{x=a} = 0 \\ u|_{y=0} = f(x) & \frac{\partial u}{\partial y}\bigg|_{y=b} = 0 \end{cases}$$

We can use the Separation of variable approach.

Step1

$$u(x, y) = X(x)Y(y)$$

According to $u_{xx} + u_{yy} = 0$, we can get,

$$\begin{aligned} X''(x)Y(y) + X(x)Y''(y) &= 0 \\ \frac{X''(x)}{X(x)} &= -\frac{Y''(y)}{Y(y)} = \lambda \\ X''(x) + \lambda X(x) &= 0 \\ Y''(y) - \lambda Y(y) &= 0 \end{aligned}$$

Step2

1

$$\lambda = 0$$

$$\begin{aligned} X''(x) &= 0 \quad X(x) = Ax + B \\ X(0) &= 0 \Rightarrow B = 0 \quad X'(a) = Aa = 0, A = 0 \\ X(x) &\equiv 0 \quad u(x, y) \equiv 0 \end{aligned}$$

which is not true

2

$$\lambda < 0$$

Suppose $\lambda = -s^2$

$$X''(x) - s^2 X(x) = 0$$

$$X(x) = Ae^{sx} + Be^{-sx}$$

$$X(0) = A + B = 0 \quad X'(a) = Ase^{sa} - Bse^{-sa} = 0$$

$$\Rightarrow -B[e^{sa} + e^{-sa}] = 0 \Rightarrow A = B = 0$$

which is not true.

3

$$\lambda > 0$$

$$X''(x) + \lambda X(x) = 0$$

$$\therefore X(x) = A \cos \sqrt{\lambda} x + B \sin \sqrt{\lambda} x$$

$$X(0) = 0 \Rightarrow A = 0$$

$$X'(x) = B \sqrt{\lambda} \sin \sqrt{\lambda} x \Rightarrow X'(a) = B \sqrt{\lambda} \sin \sqrt{\lambda} \cdot a = 0$$

$$\Rightarrow \sin \sqrt{\lambda} a = 0$$

$$\sqrt{\lambda} a = \frac{\pi(2n+1)}{2}$$

$$\sqrt{\lambda} = \frac{2n+1}{2a} \pi \quad n = 0, 1, 2, \dots$$

$$X_n(x) = \sin \frac{2n+1}{2a} \pi x$$

Step 3

$$Y''(y) - \lambda Y(y) = 0$$

$$u|_{y=0} = f(x) \quad \left. \frac{\partial u}{\partial y} \right|_{y=b} = 0$$

$$Y(y) = C e^{\sqrt{\lambda} y} + D e^{-\sqrt{\lambda} y}$$

$$Y_n(y) = C_n e^{\frac{2n+1}{2a} \pi y} + D_n e^{-\frac{2n+1}{2a} \pi y} \quad n = 0, 1, 2, 3, \dots$$

$$u_n(x, y) = X_n(x) Y_n(y)$$

$$\begin{cases} u_{xx} + u_{yy} = 0 \\ u_{x=0} = 0 \quad \left. \frac{\partial u}{\partial x} \right|_{x=a} = 0 \\ u_{y=0} = f(x) \quad \left. \frac{\partial u}{\partial y} \right|_{y=b} = 0 \end{cases}$$

Then we can get,

$$\begin{aligned}
u_n(x, y) &= \sin \frac{2n+1}{2a} \pi x \left[C_n e^{\frac{2n+1}{2a} \pi y} + D_n e^{-\frac{2n+1}{2a} \pi y} \right] \\
u|_{y=0} &= f(x) \Rightarrow \sum_{n=0}^{\infty} \sin \frac{2n+1}{2a} \pi x [C_n + D_n] = f(x) \\
f(x) &= \sum_{n=0}^{\infty} [C_n + D_n] \sin \frac{2n+1}{2a} \pi x \\
f(x) &= \sum_{n=0}^{\infty} P_n \varphi_n(x) \\
\int_0^{T_0} \varphi_n(x) \varphi_m(x) dx &= \begin{cases} 0 & n \neq m \\ E & n = m \end{cases} \\
\int_0^{T_0} f(x) \varphi_2(x) dx &= \sum_{n=0}^{\infty} P_n \int_0^{T_0} \varphi_n(x) \varphi_2(x) dx \\
&= P_2 E \\
P_n &= \frac{1}{E} \int_0^{T_0} f(x) \cdot \varphi_n(x) dx
\end{aligned}$$

Then according to

$$u_n(x, y) = \sin \frac{2n+1}{2a} \pi x \left[C_n e^{\frac{2n+1}{2a} \pi y} + D_n e^{-\frac{2n+1}{2a} \pi y} \right]$$

We can get,

$$u(x, y) = \sum_{n=0}^{\infty} \sin \frac{2n+1}{2a} \pi x \left[C_n e^{\frac{2n+1}{2a} \pi y} + D_n e^{-\frac{2n+1}{2a} \pi y} \right]$$

Then,

$$C_n + D_n = \frac{2}{a} \int_0^a f(x) \sin \frac{2n+1}{2a} \pi x dx$$

We can also know,

$$u_y = \sum_{n=0}^{\infty} \sin \frac{2n+1}{2a} \pi x \left[C_n \frac{2n+1}{2a} \pi e^{\frac{2n+1}{2a} \pi b} - D_n \frac{2n+1}{2a} \pi e^{-\frac{2n+1}{2a} \pi b} \right] = 0$$

Then we can get,

$$C_n + D_n = \frac{2}{a} \int_0^a f(x) \sin \frac{2n+1}{2a} \pi x dx$$

$$\Rightarrow C_n = -D_n \tanh \frac{2n+1}{2a} \pi b.$$

$$D_n = \frac{2}{a} \int_0^a f(x) \sin \frac{2n+1}{2a} \pi x dx$$

S13.4 Heat transfer

$$\begin{cases} u_t - k(u_{xx} + u_{yy}) = 0 \\ u_x|_{x=0} = 0 & u_x|_{x=a} = 0 \\ u_y|_{y=0} = 0 & u_y|_{y=b} = 0 \\ u_{t=0} = \phi(x, y) \end{cases}$$

We can still use the separation of variables.

Step1

$$u(x, y, t) = T(t)X(x)Y(y)$$

$$u_t - k(u_{xx} + u_{yy}) = 0$$

$$T'XY - k(TX''Y + TXY'') = 0$$

$$\frac{1}{k}(-T'XY) + (TX''Y + TXY'') = 0$$

$$\frac{X''}{X} + \frac{Y''}{Y} - \frac{1}{k} \frac{T'}{T} = 0$$

$$\frac{x''}{x} + \frac{Y''}{Y} = \frac{1}{k} \frac{T'}{T} = -\lambda$$

$$T' + \lambda k T = 0$$

$$\frac{x''}{X} + \frac{Y''}{Y} = -\lambda$$

$$\frac{X''}{X} = -\lambda - \frac{Y''}{Y} = -\mu$$

$$X'' + \mu X = 0$$

Let $\mu - \lambda = -v$

$$\begin{cases}
Y'' + vY = 0 \\
X'' + \mu X = 0 \\
u_x|_{x=0} = 0 \quad u_x|_{x=-a} = 0 \Leftrightarrow X(x)'|_{x=0} = 0, \quad X'(x)|_{x=a} = 0 \\
Y'' + vY = 0 \\
Y(y)'|_{y=0} = 0 \quad Y(y)|_{y=a} = 0
\end{cases}$$

We now focus on the first set of equations.

Step2

$$\begin{aligned}
\mu &= 0 \quad X(x) = Ax + B \text{ or } X(x) = B \\
\mu &< 0 \quad X(x) = Ae^{ax} + Be^{-ax} \\
\mu &= -s^2, s = \sqrt{-1} \quad X = Ae^{sx} + Be^{-sx} \\
X' &= Ase^{sx} + B(-s)e^{-sx} \\
&\Rightarrow \begin{cases} A - B = 0 \\ Ae^{sa} - Be^{-sa} = 0 \end{cases} \Rightarrow A = B = 0
\end{aligned}$$

With the same reason, it must be $\mu > 0$

$$\begin{aligned}
x'' + \mu x &= 0 \\
x &= A\cos\sqrt{\mu}x + B\sin\sqrt{\mu}x \\
x' &= -A\sqrt{\mu}\sin\sqrt{\mu}x + B\sqrt{\mu}\cos\sqrt{\mu}x \\
x' = 0 &\Rightarrow B = 0 \Rightarrow x = A\cos\sqrt{\mu}x \\
x' &= -A\sqrt{\mu}\sin\sqrt{\mu}x \\
x'(a) &= -A\sqrt{\mu}\sin\sqrt{\mu}a = 0
\end{aligned}$$

Then $\sin\sqrt{\mu}a = 0$

$$\begin{aligned}
\therefore \sqrt{\mu}a &= n\pi \quad n = 1, 2, 3, \dots \\
\sqrt{\mu} &= \frac{n\pi}{a} \quad X(x) = A_n \cos \frac{n\pi}{a} x \quad n = 1, 2, 3, \dots \\
\sqrt{v} &= \frac{n\pi}{b} \quad Y(y) = B_n \cos \frac{n\pi}{b} y \quad m = 1, 2, 3, \dots \\
T' + \lambda kT &= 0 \\
-v &= \mu - \lambda \Rightarrow \lambda = \mu + v = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2
\end{aligned}$$

Suppose $e^{-\lambda kt}$

$$-\lambda k e^{-\lambda t} + \lambda k e^{-\lambda k t} = 0$$

$$T(t) = e^{-\lambda k t}$$

$$u(x, y, t) = \sum T \cdot X \cdot Y = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \cos \frac{n\pi}{a} x \cos \frac{m\pi}{b} y e^{-k \left[\left(\frac{n\pi}{a} \right)^2 + \left(\frac{m\pi}{b} \right)^2 \right] t}$$

$$\phi(x, y) = u|_{t=0}^{\infty} = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} C_{n,m} \phi_{n,m}(x, y)$$

$$C_{n,m} = \int_0^a \int_0^b \phi(x, y) \cos \frac{n\pi}{a} x \cos \frac{m\pi}{b} y dx dy$$

S13.5

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t) \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = 0 \\ u_t|_{t=0} = 0 \end{cases}$$

Find $v(x, t)$ a particular solution.

$$\begin{cases} V_{tt} - a^2 V_{xx} = f(x, t) \\ V|_{x=0} = 0 \quad V|_{x=l} = 0 \end{cases}$$

Let $u(x, t) = v(x, t) + w(x, t)$

$$\begin{cases} W_{tt} - a^2 W_{xx} = 0 \\ w|_{x=0} = w|_{x=l} = 0 \\ w|_{t=0} - w|_{t=0} - v|_{t=0} = -v|_{t=0} = \varphi(x) \\ w_t|_{t=0} = u_t|_{t=0} - v_t|_{t=0} = -v_t|_{t=0} = \psi(x) \end{cases}$$

Example

$$\begin{cases} u_{tt} - a^2 u_{xx} = A_0 \sin \omega t \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = u_t|_{t=0} = 0 \end{cases}$$

$$\begin{cases} V_{tt} - a^2 V_{xx} = A_0 \sin \omega t \\ V|_{x=0} = V|_{x=l} = 0 \end{cases}$$

Assume $V(x, t) = f(x) \sin \omega t$

$$f(x)(-w^2)\sin wt - a^2 f''(x)\sin wt = A_0 \sin wt$$

$$f''(x) + \frac{w^2}{a^2} f(x) = -\frac{A_0}{a^2}$$

$$f''(x) + \frac{w^2}{a^2} f(x) = 0$$

$$f_p(x) = C, \frac{w^2}{a^2} C = -\frac{A_0}{a^2}, c = -\frac{A_0}{w^2}$$

$$f(x) = A \cos \frac{w}{a} x + B \sin \frac{w}{a} x - \frac{A_0}{w^2}$$

And we have,

$$\begin{cases} f(0) = 0 \\ f(l) = 0 \end{cases} \Rightarrow \begin{cases} A = \frac{A_0}{w^2} \\ B = \frac{A_0}{w^2} \tan \frac{wl}{2a} \end{cases}$$

Then,

$$u(x, t) = v(x, t) + w(x, t)$$

$$w = u - v$$

$$\begin{cases} W_{tt} - a^2 W_{xx} = 0 \\ W|_{x=0} = w|_{x=1} = 0 \\ W|_{t=0} = -V(x, 0) = 0 = \phi(x) \\ W_t|_{t=0} = -V_t(x, 0) = -wf(x) = \psi(x) \end{cases}$$

Then we can get,

$$\begin{aligned} W(x, t) &= \sum_{n=1}^{\infty} \left(C_n \sin \frac{n\pi}{l} at + D_n \cos \frac{n\pi}{l} at \right) \sin \frac{n\pi}{l} x \\ C_n &= \frac{2}{n\pi a} \int_0^L \psi(x) \sin \frac{n\pi}{l} x dx \\ D_n &= \frac{2}{l} \int_0^L \phi(x) \sin \frac{n\pi}{l} x dx = 0 \\ C_n &= \frac{2}{n\pi a} \int_0^l -w \left[A \cos \frac{n\pi}{a} x + B \sin \frac{n\pi}{a} x + \frac{-A_0}{w^2} \right] \sin \frac{n\pi}{c} x dx \\ &= \frac{-2A_0 w l^3}{\pi^2 a} \cdot \frac{1 - (-1)^n}{n^2} \cdot \frac{1}{(n\pi a)^2 - (wl)^2} \end{aligned}$$

S13.6

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 \\ u_{x=0} = \mu(t) \\ u_t|_{t=0} = 0 \end{cases} \quad u|_{x=l} = v(t) \quad u|_{t=0} = 0$$

$$u(x, t) = v(x, t) + w(x, t)$$

$$\begin{cases} V_{tt} - a^2 V_{xx} = 0 \\ V|_{x=0} = \mu(t) \\ V|_{t=0} = V(t) \end{cases}$$

Suppose $w = u - v$

$$\begin{cases} w_{tt} - a^2 w_{xx} = 0 \\ w|_{x=0} = w|_{x=l} = 0 \\ w|_{t=0} = -v(x, 0) \end{cases} \quad w_t|_{t=0} = -v_t(x, 0)$$

We can get,

$$V(x, t) = \frac{1}{l} [v(t) - \mu(t)] x^2 + \mu(t)$$